

Loïc Dubois

Born in 1999
French citizen
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Research interests

Graph algorithms, computational and combinatorial geometry and topology, topological data analysis.
General interests in geometry processing, combinatorial optimization, and AI.

Education

Oct. 2025-	Postdoc. University of Notre Dame (USA). Supervised by Erin Chambers.
2022-2025	Ph.D. Université Gustave Eiffel (Champs-sur-Marne, France). Supervised by Éric Colin de Verdière and Vincent Despré.
2021-2022	Projet Long de Recherche (PLR, advanced graduate class). École Normale Supérieure de Lyon (Lyon, France).
2019-2021	Master, Theoretical Computer Science. École Normale Supérieure de Lyon (Lyon, France).
2018-2019	Bachelor, Theoretical Computer Science. École Normale Supérieure de Lyon (Lyon, France).
	Bachelor, Mathematics for Engineering. Université Claude Bernard (Lyon, France).
2016-2018	CPGE (preparation for national competitive exams). Lycée Fénélon (Paris, France).

Research internships

2021-2022	PLR internship. 4 months. At Université Gustave Eiffel (Champs-sur-Marne, France). Supervised by Éric Colin de Verdière and Vincent Despré. Subject: untangling graphs on surfaces. <i>Leading to [A1, A2, A3].</i>
	PLR internship. 3 months. At Technische Universität Berlin (Berlin, Germany). Supervised by Stefan Felsner. Subject: extremal trees for the Wiener index.
2021	Master 2 internship. 2 months. At INRIA Nancy (Nancy, France). Supervised by Vincent Despré and Monique Teillaud. Subject: Delaunay flips algorithms on hyperbolic surfaces. <i>Leading to [A4, A6][B1].</i>
2020	Master 1 internship. 2 months. At Universitat Politècnica de Catalunya, remote due to covid. Supervised by Guillem Perarnau. Subject: centered colorings of graphs. <i>Leading to [A5].</i>
2019	Bachelor internship. 6 weeks. At ONERA Toulouse (Toulouse, France). Supervised by Pierre-Loïc Garoche. Subject: approximation by semi-algebraic sets.

Publications

- [A1] Éric Colin de Verdière, Vincent Despre, and Loïc Dubois. Untangling graphs on surfaces. *Journal of the ACM*, 2026.
- [A2] Éric Colin de Verdière, Vincent Despre, and Loïc Dubois. A discrete analog of tutte’s barycentric embeddings on surfaces. *TheoretiCS*, 5, 2026.
- [A3] Loïc Dubois. Making multicurves cross minimally on surfaces. In *32nd Annual European Symposium on Algorithms (ESA 2024)*, volume 308, pages 50:1–50:15, 2024. doi:10.4230/LIPIcs.ESA.2024.50.
- [A4] Loïc Dubois. Computing the Intrinsic Delaunay Triangulation of a Closed Polyhedral Surface. In Hee-Kap Ahn, Michael Hoffmann, and Amir Nayyeri, editors, *42nd International Symposium on Computational Geometry (SoCG 2026)*, volume 367 of *Leibniz International Proceedings in Informatics (LIPIcs)*, pages 40:1–40:15, Dagstuhl, Germany, 2026. Schloss Dagstuhl – Leibniz-Zentrum für Informatik. URL: <https://drops.dagstuhl.de/entities/document/10.4230/LIPIcs.SocG.2026.40>, doi:10.4230/LIPIcs.SocG.2026.40.
- [A5] Loïc Dubois, Gwenaél Joret, Guillem Perarnau, Marcin Pilipczuk, and François Pitois. Two lower bounds for p -centered colorings. *Discrete Mathematics & Theoretical Computer Science*, 22(Graph Theory), 2020. doi:10.23638/DMTCS-22-4-9.
- [A6] Loïc Dubois. A bound for Delaunay flip algorithms on flat tori. *Computing in Geometry and Topology*, 2(2):6:1–6:13, 2023. doi:10.57717/cgt.v2i2.27.

Implementation

- [B1] Vincent Despre, Loïc Dubois, Marc Pouget, and Monique Teillaud. 2D triangulations on hyperbolic surfaces. In *CGAL User and Reference Manual*. CGAL Editorial Board, 6.1 edition, 2025. URL: <https://doc.cgal.org/6.1/Manual/packages.html#PkgHyperbolicSurfaceTriangulation2>.

Reviewing

Symposium on Discrete Algorithms (SODA) – Symposium on Computational Geometry (SoCG) – European Symposium on Algorithms (ESA) – Graph Drawing (GD) – Symposium on Theoretical Aspects of Computer Science (STACS) – Discrete and Computational Geometry (DCG) – Journal of Computational Geometry (JoCG) – Computing in Geometry and Topology (CGT)

Teaching

Assembler – OpenGL – SQL – Algorithms and programming in Python – Maths for computer scientists

Languages

Python – C++ – L^AT_EX